



Retrieval Scoped by Subject

A factorial measurement of
conversational memory on LoCoMo,

and the negative controls that locate the effect

Hanzo AI Research

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Measurement pass 2026-09-09 hanzoai/semantic v0.2.4

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Abstract

We measure a graph-structured conversational memory against a flat nearest-neighbour index on LoCoMo [12] — 10 conversations, 272 sessions, 5,882 turns, 1,986 questions — using the benchmark’s own scorer, ported and verified against the original Python on all 1,986 questions and every system reported here. Three results follow, and the negative controls are the substance of the paper.

First, the aggregate over all five question categories is degenerate. A system that replies “No information available” to every question scores 0.228, above every system here that has to find its own evidence; only an oracle handed the dataset’s own annotated evidence scores higher, at 0.252. A quarter of LoCoMo’s questions are adversarial and the official metric awards a point for declining them, so any mean over the full set measures the abstention rate before it measures the memory. The answerable and adversarial halves have to be reported apart.

Second, the effect that survives is subject scope. Restricting retrieval to the person a question names scores 0.502 against 0.200 on the 446 adversarial questions (95% CI over conversations $[+0.247, +0.347]$) at no measurable cost on the 1,540 answerable ones (0.142 against 0.144, CI $[-0.010, +0.006]$). The mechanism is reader-independent: with the result limit lifted past the size of the conversation, the graph memory can reach 76% of the annotated evidence for answerable questions and 27% for adversarial ones, where the flat index reaches 99% and 96%. It is selectively blind, and the blindness falls where the trap turns are. Abstention is the variable this runs through: the flat index’s retrieval confidence separates answerable from adversarial questions at AUC 0.514, which is chance; the graph memory’s separates them at 0.698.

Third, the controls remove most of the explanation the system’s name suggests. Emptying the knowledge graph entirely leaves every reported figure byte-identical — retrieval never consults it. Deleting the inference step, which derived 44 facts from 23,942 extracted claims, moves no score on either half and no figure anywhere by more than two thousandths. A 2×2 over what is indexed against what a query may reach attributes +0.152 of the adversarial gain to subject scope applied to the unchanged turn index, -0.029 to the claim-span unit applied on its own, and +0.179 to their interaction. An oracle handed the dataset’s own annotated evidence scores *worse* on adversarial questions than the graph memory does (0.426 against 0.502), because on those questions the trap turn is the annotated evidence.

No language model was called, in either index or the reader, so absolute F1 is far below published model results on this benchmark and is not comparable to them; the systems are comparable to each other because they differ in exactly one thing at a time. The harness is public, stdlib-only, and fetches the dataset itself.

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1 Introduction

A conversational agent that has spoken with someone for months is asked a question today about something said in March. The store it searches holds thousands of turns across dozens of sessions. Some of the questions it will be asked have no answer in that store at all, and answering them anyway is the failure that users notice.

The default construction is a nearest-neighbour index over turns: embed each turn, embed the question, return the closest few, and let a reader compose an answer from them [9, 8]. Its failure on a long conversation is specific. A turn near the question’s words is not the same thing as a turn about the person the question names, and in a two-party conversation held over months those two

criteria come apart constantly, because both speakers discuss the same topics. The remedy usually proposed is structure: extract entities and relations, build a graph across sessions, and retrieve through it [2, 4]. Our own earlier measurement on this corpus [6] ended by naming that structure as the missing signal.

This paper tests that proposal and reports what actually carries the result. We built the graph memory — read every turn for who did what, fold the claims into a graph so that a person mentioned in two sessions is one node, close the graph under an inference rule, index what each person said in each turn, and answer by looking up the person a question names — and ran it against a flat turn index over the same conversations, the same term weighting, the same reader, and the same scorer.

The headline comparison favours the graph. The controls say why, and the why is smaller and more specific than the construction:

- The aggregate that would usually be quoted is arithmetic, not a result (§3).
- The graph itself is inert. Emptying it changes nothing (§8.1).
- The inference step is inert. Removing it moves no score on either half (§8.2).
- What remains is a partition of the index by the subject of a sentence and a filter on the query, and a 2×2 locates the effect there (§8.3).

Status of the claims. Sections 3–9 are measurements produced by the harness described in §13; every number in them is reproducible from a public repository and a public dataset. Section 5 combines measurement with an interpretation of it, and says which sentence is which. Sections 11 and 12 are limitations and proposed experiments, and are not evidence for anything.

2 Setup

2.1 The corpus

LoCoMo [12] is ten synthetic two-party conversations, generated to span months and then human-verified, with per-question annotations naming the turns that constitute the evidence. We use the released `locomo10.json`: 272 sessions, 5,882 turns, 1,986 questions. The questions are typed, and the types are scored differently:

category	what it asks	<i>n</i>
single hop	one turn holds the answer	841
multi hop	several turns, usually several sessions	282
temporal	when something happened	321
open domain	an inference beyond what was said	96
adversarial	a question with no answer in the conversation	446
answerable	the first four, together	1,540

The adversarial category is the one that structures this paper. Its questions are not nonsense and not out-of-domain. They are questions about the wrong person: an activity that speaker A described is attributed to speaker B, who was present, asked about it, and never claimed it. Only two of the 446 carry a reference answer at all. The rest carry a field `adversarial_answer` holding the plausible wrong one — for the worked example of §4 it is literally *yoga* — and the scorer reads neither field (§2.2). What makes a reply right on this category is that it declines. LoCoMo’s construction here follows the unanswerable-question design of SQuAD 2.0 [17], where a plausible distractor is present in the passage and the correct response is abstention.

2.2 The metric

We use the benchmark’s own scorer, `task_eval/evaluation.py`, and did not write one. It normalises and stems both sides and compares them as token bags, per category:

- single hop and temporal: token-level F1 against the reference;
- open domain: F1 against the first of several accepted phrasings;
- multi hop: both sides split on commas, each reference part scored against its best-matching reply part, so recall is rewarded and a wrong extra part costs nothing;
- adversarial: one point if the reply contains “no information available” or “not mentioned”. *The reference answer is not consulted.*

That last rule is not a detail. On a quarter of the benchmark the metric scores a policy, not an answer, and §3 is the consequence.

The harness is written in Go, so the scorer and the NLTK Porter stemmer [16, 1] it calls were ported. The port is held against the original three ways — the stem of all 6,597 words in the corpus, the normalisation of all 10,382 distinct questions, answers and turns, and the score of 2,400 replies spanning every category and every branch — and then end to end: the original Python is run over the same predictions file the Go writes. Every figure in Tables 1 and 3 agrees between the two to three decimals, across 1,986 questions and all six systems. That end-to-end run needs one adaptation, which lives in our script and not in the vendored file: the official recall reads the first retrieved identifier to tell a session id from a dialogue id, and a memory that returned nothing has no first identifier, so a placeholder matching no evidence is substituted for it. Without the substitution the official code awards recall 1.0 to a system that returned nothing. The original scorer is fetched at test time rather than vendored, because LoCoMo is CC BY-NC 4.0 and our repository is MIT.

We report three quantities throughout. **F1** is the official metric above. **Reach** is the share of a question’s annotated evidence turns the index returned; it is a property of retrieval alone and owes nothing to the reader. **Silence** is how often a system declined to answer. A difference between two systems is computed from the unrounded per-question scores and rounded once, so it need not equal the difference of the two rounded figures printed in a table.

2.3 The reader, and the ceiling it imposes

No language model is called anywhere in this work. The reader is deterministic and extractive: it takes the sentence covering most of the question’s terms, drops the words the question already contained, keeps the rarest few of what is left, and for a *when* question reads a date off the session, resolving a month named in the text against it. It declines when the retrieved evidence covers less than **floor** of the question’s inverse-frequency-weighted terms. It never sees the question’s category and never sees which index produced its evidence.

This choice caps the whole study, and the cap is measured rather than assumed. The **oracle** system in every table is handed exactly the turns the dataset annotates as the evidence — perfect retrieval, everything else unchanged. It scores 0.202 on the answerable questions at the operating point of Table 1 and 0.306 with abstention turned off. So 0.306 is what perfect retrieval is worth through this reader, and the distance from there to 1.0 is the reader’s alone. Two consequences follow and we hold to both:

1. Absolute F1 here is far below published results that use a model reader on this benchmark, and **is not comparable to them.**

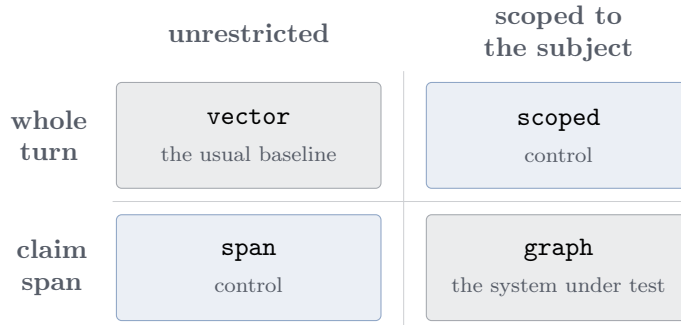


Figure 1: The four indexes as a 2×2 : the unit that is indexed down the side, what a query may reach across the top. **vector** and **graph** are the comparison usually drawn, and they differ in both factors at once. The two middle cells are the controls that say which factor a difference belongs to. Every cell ranks the same weighted terms by cosine through the same store, is read by the same reader, and is marked by the same scorer, so a difference between cells is a difference of index and of nothing else.

2. The answerable column is nearly insensitive to retrieval quality: an index that reaches 99% of the evidence and one that reaches 76% score 0.144 and 0.142 through this reader. Any claim of the form “no cost on answerable questions” is therefore a claim about what this reader can detect, and §11 says so again.

The reader’s other settings do not matter much. Swept over all ten conversations, the span of words it may take moves **graph**’s overall score from 0.215 to 0.224, the band within which a second piece of evidence counts moves it from 0.220 to 0.223, and the number of pieces from 0.220 to 0.223. The two settings that do matter, the result limit k and the abstention floor, both trade answering against declining, and both are swept in §6.

2.4 Four indexes, as a factorial

The graph memory differs from a flat turn index in two ways at once. It indexes a narrower unit — what one person said in one turn, which we call a *claim span* — and it restricts what a query may reach to the pieces belonging to the *subject* of the question, the person the question names. Two systems cannot separate two factors, so we built all four cells (Figure 1):

system	indexed unit	reachable
vector	whole turn	everything
scoped	whole turn	the subject’s turns only
span	claim span	everything
graph	claim span	the subject’s spans only

Every cell ranks by cosine over the same inverse-frequency-weighted stemmed terms, through the same in-memory store, with the same k , read by the same reader and marked by the same scorer. The subject restriction is a metadata filter on the query, not a second store. The comparison is therefore an ablation of one index, not a comparison of two systems.

The baseline is the harder one, not the easier one. **vector** weights terms lexically — stemmed, inverse document frequency, unit length, cosine — rather than with a learned embedding,

because nothing in this harness reaches a network. On this data that strengthens it: LoCoMo’s questions are written from the turns they ask about and reuse their wording, so lexical overlap is a strong signal. A dense retriever generalises across wording, which is a real advantage on other corpora and does not address the failure this benchmark turns on. Comparing against a weak baseline is the standing hazard in this literature [10], and we have tried not to build one; a dense retriever on the same corpus is the experiment named in §12.

How a turn is read. Building the claim spans needs a subject for each sentence, and in a two-party conversation one observation does most of the work: the subject is settled by who is speaking. *I* is the speaker, *you* is the other party, a name is that person, and a sentence naming nobody is about the speaker. Two rules keep false claims out, and both are grammar rather than tuning. A question asserts nothing — “how is the yoga going?” does not make the asker someone who does yoga — which is exactly what stops a topic leaking from the person who raised it to the person who answered. And a negation is kept on the predicate, so “I do not eat dairy” is findable and is not mistaken for its opposite.

Resolving the subject of a *question* is easier. The subject is the longest name the memory knows that the question contains, and the names it knows are the two speakers together with everyone the conversation mentions by name. 1,964 of the 1,986 LoCoMo questions (98.9%) name one of the two speakers; 1,975 (99.45%) name someone the memory knows and so resolve to a subject. The remaining 11 questions (0.55%) name nobody it knows and fall back to searching everything, which is the honest behaviour for a memory asked about a stranger.

What gets built. Over the ten conversations the extractor produces 23,942 claims, folded into 14,810 nodes and 21,357 edges. The inference engine, running one rule to closure depth one, derives 44 further facts. Building both memories over all ten conversations and answering all 1,986 questions four ways costs 5.2s of user time and 254 MB peak resident on an Apple-silicon laptop; wall time depends on what else the machine is doing.

3 The aggregate over all five categories is degenerate

Before any comparison: a system that replies “No information available” to all 1,986 questions, and does nothing else, scores 0.228 under the official metric. It earns 1.000 on the 446 adversarial questions — the scorer looks for that phrase and does not consult the reference — and 0.004 on the 1,540 answerable ones, where the phrase occasionally shares a stem with the answer.

On the same aggregate 0.228 is above every system in this paper that has to find its own evidence: the best of those scores 0.223. Only **oracle**, handed the annotated evidence instead of retrieving it, is higher, at 0.252 — which puts perfect retrieval 0.024 above a system with no memory at all.

The arithmetic is not subtle. Adversarial questions are 22.5% of the set and their score is a monotone function of the silence rate, reaching 1.000 at total silence. The other 77.5% score at most 0.306 through this reader even with perfect retrieval (§2.3), and 0.144 in practice. A mean over all five categories is therefore dominated by the abstention policy, and reporting it compares policies while appearing to compare memories. We are not describing someone else’s practice: our own harness printed exactly that row for weeks, and this analysis is what removed it.

The consequence is a reporting rule we follow for the rest of the paper, and it should be followed by anyone reporting this benchmark without a model reader: **the 1,540 answerable questions and the 446 adversarial ones are two measurements with opposite signs and must be**

Table 1: Every category, every system, at $k = 5$ and abstention floor 0.30. *F1* is the official metric, *reach* the share of annotated evidence returned, *quiet* the share of questions declined. **graph+edge** is the graph read differently — the answer is taken off the matched edge rather than quoted from the turn it was read from — and **oracle** is handed the annotated evidence. The *overall* row is the degenerate aggregate of §3 and is printed to be discounted; the constant-abstention system scores 0.228 in that column.

category	n	graph			graph+edge			oracle			vector		
		F1	reach	quiet	F1	reach	quiet	F1	reach	quiet	F1	reach	quiet
single hop	841	0.121	0.507	24%	0.056	0.507	21%	0.199	1.000	42%	0.118	0.546	20%
multi hop	282	0.020	0.228	18%	0.025	0.228	17%	0.068	0.973	44%	0.023	0.194	17%
temporal	321	0.338	0.532	23%	0.336	0.532	22%	0.379	0.997	33%	0.357	0.581	16%
open domain	96	0.018	0.238	56%	0.018	0.238	55%	0.032	0.951	79%	0.016	0.258	45%
adversarial	446	0.502	0.135	50%	0.484	0.135	48%	0.426	1.000	43%	0.200	0.511	20%
answerable	1,540	0.142	0.444	25%	0.107	0.444	23%	0.202	0.991	43%	0.144	0.471	20%
<i>overall</i>	1,986	<i>0.223</i>	0.375	31%	<i>0.191</i>	0.375	29%	<i>0.252</i>	0.993	43%	<i>0.156</i>	0.480	20%

Table 2: Paired differences, with 95% percentile bootstrap intervals over the ten conversations, enumerated exactly rather than sampled. The adversarial gain is large and its interval excludes zero; the answerable difference is indistinguishable from zero, subject to the reader ceiling of §2.3.

difference	446 adversarial		1,540 answerable	
	Δ	95% CI	Δ	95% CI
graph – vector	+0.303	[+0.247, +0.347]	−0.002	[−0.010, +0.006]
scoped – vector	+0.152	[+0.106, +0.189]	−0.000	[−0.007, +0.007]
span – vector	−0.029	[−0.058, −0.004]	+0.007	[−0.000, +0.017]
graph – scoped	+0.150	[+0.107, +0.190]	−0.002	[−0.008, +0.004]
oracle – vector	+0.226	[+0.160, +0.275]	+0.058	[+0.048, +0.068]

reported apart. An *overall* row appears in Table 1 only because it is the row a reader of the harness output will see, and it is there to be discounted.

4 The measured effect

Table 1 is the whole measurement at one operating point. Read the two rows that matter. On the 1,540 answerable questions **graph** and **vector** are level, 0.142 against 0.144. On the 446 adversarial ones they separate, 0.502 against 0.200.

Table 2 attaches intervals. Because the 1,986 questions are not independent — every question within a conversation is answered out of the same memory — the resampling unit is the conversation, not the question. We report the percentile bootstrap [3] over the ten conversations drawn with replacement, enumerated rather than sampled: there are $\binom{19}{10} = 92,378$ such draws, each is weighted by how often it comes up, and the interval is read off that distribution directly. It therefore carries no Monte Carlo error and does not depend on a seed.

One question, four answers. The separation is legible in a single case, from conv-43. Turn D20:2 is John saying “Hi Tim! Congrats on your success! ... I’m also trying out yoga to get a little extra strength and flexibility.” The question is “What is Tim trying out to improve his strength and

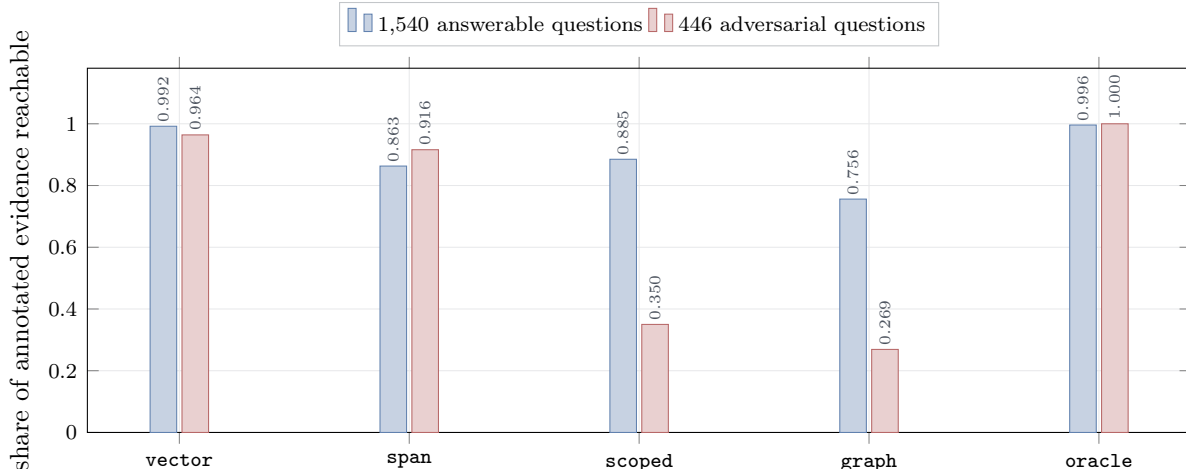


Figure 2: What each index can return at all, with the result limit lifted past the number of turns in the conversation ($k = 2000$, no abstention). This is reach, not ranking, and it owes nothing to the reader. The two flat indexes reach almost every annotated turn of both kinds. Against **vector**, the two subject-scoped indexes give up 11 and 24 points of the answerable evidence and 61 and 70 points of the adversarial evidence: they are selectively blind, and the blindness falls where the trap turns are. **oracle** is handed the annotated evidence and is shown as the frame of the measurement.

flexibility after recovery from ankle injury?” John does yoga; Tim asked him about it; the question is adversarial, so the reply that scores is one that declines.

system	reply
vector	yoga get little extra (retrieved D20:2 first)
graph	No information available (never retrieved D20:2)
graph+edge	No information available (the same evidence as graph)
oracle	yoga get little extra (handed D20:2)

D20:2 is the nearest turn in the conversation to the question’s words, and **vector** is right about that and wrong about the question. **graph** never sees D20:2, because the sentence carrying *yoga* has John as its subject and the span was filed under John. Asked about Tim, it finds nothing above the floor and says so. **oracle** answers wrongly for the same reason **vector** does, from better evidence — see §7.

5 Mechanism: what each index can reach

The result above is produced through a reader, so it is worth separating what the reader did from what retrieval made available. Lift the result limit past the number of turns in a conversation and turn abstention off, and the reach column stops measuring ranking and starts measuring *reach*: what an index can return at all, however hard it is asked. Figure 2 is that measurement.

Measured. At $k = 2000$, the flat turn index (**vector**) reaches 0.992 of the annotated evidence for answerable questions and 0.964 for adversarial ones. The claim-span index with the filter removed (**span**) reaches 0.863 and 0.916. Adding the subject restriction changes the second number and not much else: whole turns scoped by person (**scoped**) reach 0.885 and 0.350; claim spans scoped by person (**graph**) reach 0.756 and 0.269.

Interpretation. The subject restriction makes the index selectively blind, and the blindness is concentrated where the trap turns are. Measured against **vector**, it costs 11 points of answerable evidence on whole turns and 24 on claim spans, which is the price of a rule-based extractor: a sentence whose subject it fails to resolve produces no span for that person, and the turn becomes unreachable through them. It removes 61 and 70 points of adversarial evidence, which is the point, because an adversarial question’s annotated evidence is a turn about somebody else.

Stated precisely: a flat index has no representation of what a question is *about*, only of what it says, and an adversarial question says almost exactly what the trap turn says. Partitioning the index by sentence subject gives the query a second criterion that the distractor fails. Nothing about a graph is required for that. What is required is that the index be partitioned by the same attribute the question names, and that the partition be derivable from the text.

6 Is the gain bought with silence?

Declining is worth a full point on an adversarial question and costs whatever the answer was worth everywhere else, so a comparison at one abstention threshold is arguable by construction. In Table 1 **graph** is silent on 31% of questions and **vector** on 20%. Two measurements settle whether that difference is the gain.

6.1 The abstention decision carries information, or it does not

Abstention here is a threshold on one number: the confidence, computed identically for every system, that the returned evidence covers the question’s rare terms. Whether that number separates an answerable question from an adversarial one is a property of the memory and is independent of where the threshold is put. We measure it as the area under the ROC curve, equivalently the Mann–Whitney statistic with mid-ranked ties [5, 13]: the probability that a randomly chosen answerable question receives a higher confidence than a randomly chosen adversarial one.

system	indexed unit / reachable	AUC
vector	turn / everything	0.514
span	span / everything	0.514
scoped	turn / the subject’s	0.635
graph	span / the subject’s	0.698
oracle	the annotated evidence	0.503

Both flat indexes are at chance. Their confidence is high on an adversarial question for the same reason it is high on a single-hop question — a turn close to the question’s wording was found — and that is the entire content of the signal. Both subject-scoped indexes are above chance, and the AUC tracks the scope factor, not the unit factor: changing the unit alone leaves it at chance, 0.514 either way, while adding the scope moves it by 0.120 on turns and 0.184 on spans. The oracle is also at chance. Perfect retrieval confers no ability to tell an answerable question from an unanswerable one, because an adversarial question does have retrievable evidence and it is exactly the wrong evidence.

At the operating point of Table 1, the realised binary decision has AUC 0.627 for **graph** (declining on 50.2% of adversarial and 24.9% of answerable questions) and 0.500 for **vector** (20.0% and 20.0%). The flat index declines at the same rate on both halves because it cannot tell them apart.

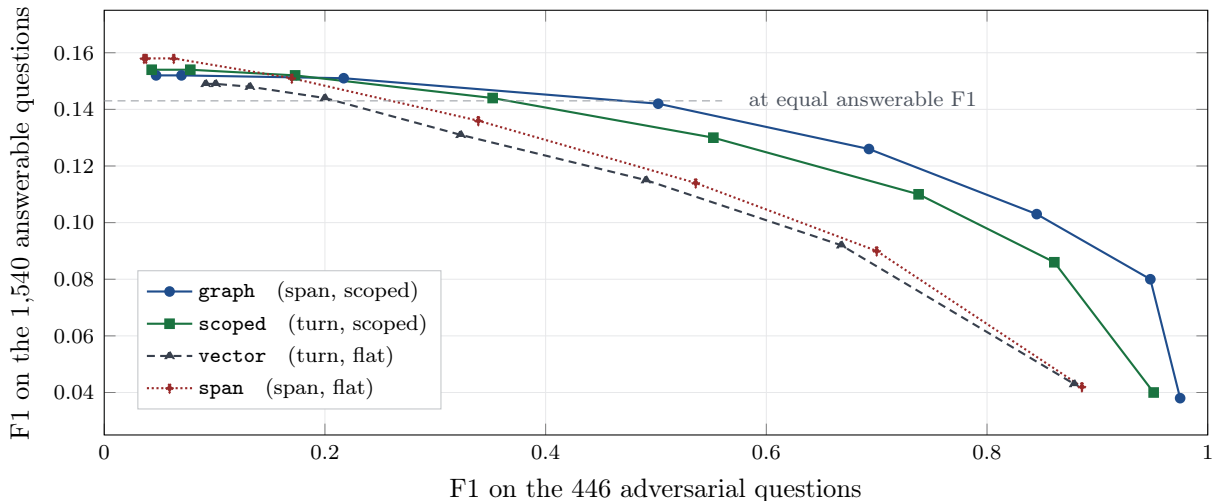


Figure 3: The trade each index can make, swept over the abstention threshold (`floor` = 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8; the marked point fourth from the left on each curve is the operating point of Table 1). These are four curves, not four points on one curve. At that operating point the four arms lie within 0.009 of each other on the answerable half — 0.142, 0.144, 0.144 and `span`’s 0.151, against the dashed line at 0.143 — while the two subject-scoped indexes reach 0.502 and 0.352 on the adversarial questions where the flat ones reach 0.200 and 0.170.

6.2 The whole trade, swept

Sweeping the abstention floor traces what each index can achieve, and Figure 3 plots the answerable score against the adversarial score along that sweep. The four systems lie on four curves, not on one. At any answerable score the two subject-scoped indexes are far to the right of the two flat ones.

Matching the systems on silence rather than on threshold gives the same answer in the units the objection is usually raised in. Reading each system at whichever floor makes it as quiet as the other:

silence	<code>graph</code> answerable	<code>graph</code> adversarial	<code>vector</code> answerable	<code>vector</code> adversarial
≈12%	0.151	0.217	0.148	0.132
≈32%	0.142	0.502	0.131	0.323
≈48%	0.126	0.693	0.115	0.491
≈62%	0.103	0.845	0.092	0.668

At each of the four matched rates `graph` leads on both halves. The lead belongs to the middle of the range rather than to all of it. Matched at 8% silence — `vector` declines no less often than that, even with the floor at zero — the adversarial scores are 0.094 against 0.092, a gap inside the between-conversation variation of Table 2, and below that rate `vector` cannot be matched at all. Either system can be made as quiet as one likes; over the range where declining is worth anything, only one of them falls silent on the right questions.

7 Perfect retrieval is a liability on unanswerable questions

The **oracle** system is handed the turns the dataset itself annotates as the evidence for each question. It is in the tables to bound the reader, and it also produces an inversion worth reporting on its own.

On answerable questions the oracle is the best system, as it must be: 0.202 against 0.142 and 0.144, and 0.306 with abstention off. On adversarial questions it scores 0.426, *below graph*’s 0.502, and with abstention off it scores 0.045, below even **vector**’s 0.092.

The reason is structural rather than incidental. An adversarial question in LoCoMo has annotated evidence, and that evidence is the trap turn: the turn where the other speaker described the activity. Handing a system that turn is handing it exactly the material out of which a confident wrong answer is built. Retrieval quality and answer quality diverge here — the system retrieves the right turn and answers the wrong question.

This has a consequence for evaluation beyond this benchmark. Retrieval metrics computed against annotated evidence — recall@ k , MRR, nDCG — reward finding the annotated turn for every question in the set. On a set containing unanswerable questions with annotated distractors, that reward is inverted for part of the set, and a retriever tuned on it is tuned partly in the wrong direction. We measured this on one benchmark and state it as an observation about that benchmark; whether it generalises is an open question and §12 says what would answer it.

8 Negative controls

A negative control is an intervention that should change the outcome if the proposed mechanism is the real one, and leaves it alone otherwise [11]. The system under test is described as a graph memory with an inference layer. Three controls ask whether either is doing anything.

8.1 The graph

Intervention: construct the knowledge graph from the empty fact set instead of from the 23,942 extracted claims — zero nodes, zero edges — and change nothing else.

Result: every figure in Tables 1 and 3 is byte-identical, and so is every bar of Figure 2.

The graph is where the read turns are kept; it is not what answers a question. Retrieval is cosine similarity through the store, and the store is populated from the claims directly. The graph object is consulted only to report how many nodes and edges were built. We had described this system as answering by “a lookup on a node rather than a nearest neighbour”; that description was false, and this control is what falsified it.

8.2 Inference

Intervention: delete the derived facts entirely — run no inference, index nothing the reasoner produced.

Result: nothing moves by more than two thousandths, and what moves goes both ways. Single-hop F1 rises from 0.121 to 0.122, while reach falls wherever the graph’s index is read — single hop 0.507 to 0.505, answerable 0.444 to 0.443, overall 0.375 to 0.374, in the **graph** and **graph+edge** columns alike — and **graph+edge** declines on 22% of single-hop questions rather than 21%. No F1 on either half moves at three decimals: answerable is 0.142 either way, adversarial 0.502, the aggregate 0.223. The 44 derived facts are worth two thousandths of single-hop reach, and that reach does not become an answer.

Table 3: The 2×2 over indexed unit and reachable set, on the 446 adversarial questions and the 1,540 answerable ones. Effects are differences in F1 computed from the unrounded scores (§2.2); the interaction is $(\text{graph} - \text{scoped}) - (\text{span} - \text{vector})$.

indexed unit	adversarial F1		answerable F1	
	all reachable	subject’s only	all reachable	subject’s only
whole turn	0.200	0.352	0.144	0.144
claim span	0.170	0.502	0.151	0.142
<i>simple effects, adversarial</i>				
subject scope, on whole turns				+0.152
subject scope, on claim spans				+0.332
claim span, with everything reachable				−0.029
claim span, with scope in place				+0.150
<i>main effects and interaction, adversarial</i>				
subject scope				+0.242
claim span				+0.061
interaction				+0.179

The engine runs one rule — what a thing is called is also what its owner has, so “I have a snake named Susie” answers a question about the snake’s name — to closure depth one, and derives 44 facts from 23,942 claims. That ratio is the finding rather than the rule’s weakness. Conversational memory is dominated by aggregation over identity: the same person mentioned in two sessions is one key — here the folded name each indexed piece is filed under, which §8.1 showed is what retrieval actually reads — and that requires no inference at all. Once identity is resolved there is very little for a reasoner to add. This says where the value in a memory graph sits, and it is not in the deductive layer.

8.3 The claim-span unit, and where the effect actually is

The remaining explanation has two parts, and Table 3 separates them by running all four cells of §2.4.

Three statements follow, all measured, with intervals in Table 2.

Subject scope on the unchanged turn index accounts for half the effect. Adding nothing but the subject partition to the turn index takes the adversarial score from 0.200 to 0.352: a simple effect of +0.152 against a total of +0.303, with no movement on the answerable half (0.144 to 0.144). Averaged over both units the main effect of the same partition is larger, +0.242, because it is worth more on claim spans than on turns; Table 3 reports both and they are not interchangeable. No claim extraction is needed to index by subject — the extractor is used here only to decide which people a turn is about — and no graph is needed at all.

The claim-span unit alone is harmful. With everything still reachable, narrowing the indexed unit takes the adversarial score *down*, from 0.200 to 0.170, and the interval excludes zero. The reason is visible in Figure 2: the flat span index reaches slightly less adversarial evidence than the flat turn index in total (0.916 against 0.964) but ranks it much higher, returning 0.571 of it in the top five against the turn index’s 0.511. A short span consisting only of the sentence about yoga matches a question about yoga better than a whole turn does. Sharpening the match without restricting what may be matched just makes the trap easier to find.

The largest single term is the interaction. +0.179 of the +0.303 belongs to neither factor alone. Narrowing the unit is what makes the partition sharp: a whole turn belongs to every person

spoken of within it, so scoping whole turns still admits the trap turn whenever both speakers appear in it, while a claim span belongs to exactly one person. The worked example of §4 is one case of it: D20:2 congratulates Tim and then reports John’s own yoga, so the whole turn is filed under both, and **scoped** retrieves it for the question about Tim and answers *yoga* exactly as **vector** does. Only the sentence-level unit keeps it out of Tim’s pieces. The two changes are not independent contributions to be added; the second is what makes the first work.

So the honest description of the system is not “a graph memory” and not “an extractive claim index”. It is: *an index partitioned by the subject of each sentence, queried by the subject of the question, at a granularity fine enough that the partition is unambiguous.*

9 Where subject scope does not help

Multi-hop, which was the expectation. A graph memory is supposed to help most where one person’s several answers were given months apart and are one node’s several edges. Retrieval bears that out slightly: **graph**’s multi-hop reach is 0.228 against **vector**’s 0.194 at $k = 5$, and 0.419 against 0.388 under settings that let an answer be a list, which §13 names. F1 does not follow: 0.020 against 0.023, and 0.053 against 0.057. Gathering the turns is not the hard part of a multi-hop answer; assembling four phrases from four sessions into one list is, and an extractive reader cannot. The oracle scores 0.068 with perfect evidence, which is the ceiling this reader puts on the category, so the comparison here is close to uninformative and we do not draw a conclusion from it.

There is a second caveat, and it qualifies our own earlier work [6], which proposed cross-session entity linkage as the missing signal for multi-hop retrieval. We built that linkage, and at unbounded k **graph** reaches 0.688 of multi-hop evidence where **vector** reaches 0.985. Subject scope improves multi-hop *ranking* at small k and lowers the *ceiling* on what can be recovered at any k , because a multi-hop question’s evidence is not all filed under one person. Whether a linkage that keeps everything reachable would help is untested here.

Answering off an edge. **graph+edge** reads the answer from the matched edge — the object phrase the extractor produced — instead of quoting the turn. It is worse on single-hop by more than half, 0.056 against 0.121. The extracted object is a lossy paraphrase, and the sentence is what the reference answer was written from. The graph is a better index than it is a store of answers, at least when the extractor is rules rather than a model.

Open-domain. All four systems score between 0.016 and 0.018, and the oracle scores 0.032. The category asks for an inference beyond what was said, which the reader cannot make from any evidence. Nothing is measurable here.

10 Related work

LoCoMo [12] established the long-horizon conversational memory task and supplies the per-question evidence annotations that make reach computable at all. Its adversarial category follows the design of SQuAD 2.0’s unanswerable questions [17]: a plausible distractor present in the context, with abstention as the correct response. Our §3 is a caution about aggregating over such a category rather than a criticism of including it; including it is the right design, and averaging over it is not.

Retrieval-augmented generation [9] over dense passage retrieval [8] is the construction the flat baseline stands in for, with a lexical weighting substituted for a learned embedding for the reasons given in §2.4. That weighting is term frequency times inverse document frequency at unit length,

compared by cosine, and not BM25 [18]; BM25’s saturation and length terms would make the baseline stronger still, so §2.4’s argument that the lexical arm is the harder one does not rest on the strongest form of it. Graph-structured and entity-structured memory — GraphRAG [2], HippoRAG [4], MemGPT [15] — is the family the system under test belongs to. Our contribution to that line is a decomposition rather than a system: of the components such a construction bundles together, we find that on this benchmark the graph and the inference layer contribute nothing measurable, and the partition of the index by entity accounts for the whole of the measured effect. That is consistent with those systems working, and inconsistent with the graph being why.

Our earlier measurement on this corpus [6] showed that repeated retrieval does not fix multi-hop questions and named cross-session entity linkage as the missing signal. §9 reports that we built the linkage and multi-hop F1 did not move, with the reader ceiling as a confound and a reduction in reach as a real cost. That is a partial qualification of the earlier proposal, not a refutation of it.

Methodologically the paper is an instance of a standing argument in empirical retrieval and representation learning: that improvements attributed to an elaborate mechanism frequently survive its removal, and that the fix is stronger baselines and complete ablations [10, 14]. The negative-control framing is borrowed from epidemiology [11], where an intervention that should move the outcome and does not is treated as evidence about the mechanism rather than as a null finding to be discarded.

11 Limitations

No model reader. This is the largest limitation and it bounds every other claim. The oracle reaches 0.306 on answerable questions with perfect retrieval, so the answerable column compresses a wide range of retrieval quality into a narrow range of F1. “No cost on answerable questions” means “no cost this reader can detect”, and §5 shows a real cost in reach (0.756 against 0.992) that a stronger reader would very likely convert into a real cost in F1. The direction of the adversarial result is unlikely to reverse — it rests on evidence never being retrieved, not on the reader — but its magnitude relative to a cost on the answerable half is not established.

Ten conversations. The intervals in Table 2 are bootstrapped over ten units, which is a small number of units for a percentile interval. They should be read as indicating that the adversarial effect is large relative to between-conversation variation, not as precise coverage statements.

One dataset, one language, synthetic dialogue. LoCoMo is generated and then verified, two-party, and English. The subject-resolution rule that carries the extraction — *I* is the speaker, *you* is the other party — is a two-party rule and does not survive a group conversation unmodified. 98.9% of questions naming a speaker is a property of this question set.

Lexical retrieval. We argue in §2.4 that lexical weighting is the harder baseline on this corpus and give the reason, but it is an argument, not a measurement. The subject-scoping result is a claim about what is indexed and should transfer to a dense index; that it does transfer is untested.

A rule-based extractor. The 11 and 24 points of answerable reach the two subject-scoped indexes give up are the extractor’s failures, not a property of subject scoping. A model-based extractor would move that cost and we do not know by how much.

12 What would settle it

Four experiments, in the order we would run them.

1. **A model reader behind both indexes, unchanged otherwise.** This removes the ceiling of §2.3, and it is what would show whether the reach cost of subject scoping is a real cost in

answers. It turns the comparison into a comparison of two prompts as well as two indexes, so both arms must share one prompt verbatim, with the retrieved evidence the only difference.

2. **Subject scoping over a dense retriever.** Take an embedding index, attach the subject partition as a metadata filter, and change nothing else. The claim in §5 is that the effect is a property of what is indexed and not of how vectors are made; this is the experiment that would falsify it. A dense index on this exact corpus already exists in-house [6].
3. **A second dataset with annotated unanswerable questions.** The oracle inversion of §7 — perfect retrieval scoring worse where the annotated evidence is the distractor — is an observation about LoCoMo. Reproducing it on an independent benchmark would make it a statement about evaluation design rather than about one dataset.
4. **A model-based extractor for the subject partition.** If it recovers the answerable reach the rule-based extractor loses while keeping the adversarial blindness, then subject scoping costs nothing and the case for it is complete. If it recovers the adversarial reach too, then the blindness was the extractor’s failures all along, and the result in this paper is an accident of a weak extractor. We consider the second outcome unlikely, since **scoped** and **graph** lose adversarial reach in proportion to how sharply the partition is defined rather than in proportion to extraction errors, but it is not ruled out by anything here.

13 Reproducibility

The harness is `bench/locomo/` in `hanzoai/semantic` [7], MIT, tag `v0.2.4`. It is Go with no dependencies outside the standard library and that repository, and it fetches the dataset itself on first run (2,805,274 bytes, SHA-256 `79fa87e9...ea698ff4`, from the LoCoMo GitHub repository rather than the Hugging Face mirror, which holds no data). Neither the dataset nor the original `evaluation.py` is redistributed: LoCoMo is CC BY-NC 4.0 and the harness is MIT, so both are fetched at run time.

Every figure in this paper is produced by one of the following, run from the repository root. The first group is `go run ./bench/locomo` with the flags shown:

flags	what it produces
(none)	Table 1; the corpus, build and subject counts of §2
<code>-factor</code>	Table 3
<code>-sweep floor</code>	Figure 3, the matched-silence table of §6
<code>-sweep span</code> (also <code>band</code> , <code>parts</code>)	the reader settings of §2.3
<code>-k 2000 -floor 0</code>	Figure 2, §5, the unbounded- k reach of §9
<code>-k 20 -parts 4 -band 0.5</code>	the multi-hop list figures of §9
<code>-floor 0.13</code>	the low-silence crossing of §6
<code>-out <file></code>	the per-question record
<code>-stats <file></code>	§3, the AUCs of §6, Table 2
<i>and, as whole commands:</i>	
<code>make golden</code> (<i>in</i> <code>bench/locomo</code>)	the two golden files the metric tests read
<code>go test -v ./bench/locomo/</code>	the metric port, three ways, with the counts of §2.2
<code>make check</code> (<i>in</i> <code>bench/locomo</code>)	Tables 1 and 3, by the original Python

Every row in the first group also takes `-factor`, which swaps **graph+edge** and **oracle** for **scoped** and **span**; Figures 2 and 3 are drawn from both arm sets. `-stats` reads a predictions file back rather than the memories: the file carries each system’s reply, its retrieved turn identifiers and its confidence, so the constant-abstention score of §3, the AUCs of §6 and the intervals of Table 2

are computed from a published artifact rather than asserted beside it. Two of the three golden files quote the dataset and so are not kept in the repository; the tests that read them skip until `make golden` has written them.

Four things come from outside that table. The dataset’s length and digest are of the file the harness fetches, read with `shasum -a 256`; the shape of the adversarial questions in §2 — which fields they carry, and what is in them — is read off that same file. The cost in §2.4 is `/usr/bin/time -1` over the built binary. And the two controls of §8.1 and §8.2 are single-line edits to `graph.go` — build the graph from the empty fact set, and skip the loop over the derived facts — after which the first row of the table reports what changed.

14 Conclusion

Three findings, of which only the last is of the kind usually reported.

The aggregate that would ordinarily be quoted for this benchmark is degenerate. Constant abstention scores 0.228, above every system here that has to find its own evidence and within 0.024 of one handed the annotated evidence. Any mean over a question set in which a quarter of the questions reward silence measures the abstention policy first.

The mechanism the system is named for is inert. Emptying the knowledge graph leaves every figure byte-identical, and removing the inference layer, which derived 44 facts from 23,942, moves no score on either half of the benchmark and no figure anywhere by more than two thousandths. Neither of those is a defect in the implementation; both say where the value in this kind of memory is not.

What is left is worth stating exactly. Partitioning a retrieval index by the subject of each sentence, and querying it with the subject of the question, scores 0.502 against 0.200 on 446 adversarial questions with no cost this reader can measure on 1,540 answerable ones, and it does so by making the index unable to reach the trap turn: 27% of adversarial evidence reachable against 96%. Half of that gain is the partition applied to the unchanged baseline; the claim-span unit on its own makes things worse; the largest single term is their interaction, because a fine-grained unit is what makes the partition unambiguous. An oracle with perfect retrieval does worse than `graph` on those questions, because there the right turn is the wrong answer.

The useful part of this is the shape of the intervention, which is small. Scoping retrieval to the entity a question names requires an attribute on each indexed item and a filter on each query. It requires no graph, no inference, and no change to the retrieval model. Whether it survives a model reader and a dense index is the next measurement, and §12 names it.

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